

Yahao Fan

+86 18539529316

fanyahao@shanghaitech.edu.cn

<https://fanyahao1.github.io/>



Education

- 2024 – Present **Master of Engineering in Electronic Information, ShanghaiTech University.**
Research Interests: *Artificial Intelligence, Embodied AI.*
- 2020 – 2024 **Bachelor of Science in Computer Science and Technology, Zhengzhou University.**

Research & Project Experience

- 2024 – Present **Research Member, YesAI Laboratory, ShanghaiTech University.**
Optimized large-scale parallel reinforcement learning pipelines for humanoid control using **Isaac Lab** and **Isaac Gym**. Executed robust policy transfer onto **Unitree G1** and **H1** robots, mitigating the sim-to-real gap through domain randomization and dynamics tuning.
- 2021 – 2023 **Research Member, Simulation Robotics Laboratory, Zhengzhou University.**
Developed reinforcement learning algorithms for multi-agent collaboration and defensive strategies in RoboCup soccer robots.

Publications

- *Fan, Y., *Gui, T., *Ji, K., Ding, S., Zhang, C., Gu, J., Yu, J., Wang, J., & Shi, Y. (2025). *DreamPolicy: A Unified World-model Policy for Scalable Humanoid Locomotion*. arXiv preprint arXiv:2505.18780.**
- Gui, T., Yang, K., **Fan, Y.**, Zhang, C., Wang, J., & Shi, Y. (2025). *Sisyphus: An attention-driven humanoid object interaction policy*. IROS 2025 Workshop: Augmentative Human–Robot Interaction.
- Liu, Z., Ji, K., Yang, K., **Fan, Y.**, Yu, J., Shi, Y., Wang, J., & Shi, Y. (2025). *Commanding humanoid by free-form language: A large language action model with unified motion vocabulary*. arXiv preprint arXiv:2511.22963.

Technical Skills

- Programming **Python, C++.**
- Algorithms **Reinforcement Learning (PPO, AMP, PULSE), ROS/ROS2, Embodied AI frameworks.**
- Simulation & Sim-to-Real **Isaac Lab, Isaac Gym, MuJoCo, Genesis. Expertise in Domain Randomization, addressing reality gaps for dynamic locomotion and tracking policies.**
- Robotic Platforms **Unitree **G1**, Unitree **H1/H1-2**, Booster Robotics **T1**, Unitree **Go2**.**

Academic Services & Community Contributions

- 2025 **ICCV 2025 Workshop Challenge Organizer.**
Contributed to the organization of the ICCV 2025 Workshop *Human-Robot-Scene Interaction and Collaboration*, specifically the *Multi-Terrain Humanoid Locomotion Challenge*.
Responsible for benchmark platform development, evaluation environment maintenance, and competition rule design. Assisted in building an Isaac Gym-based multi-terrain humanoid locomotion benchmark, and contributed to public evaluation and community organization efforts.
- Global Developer Pioneer Summit (GDPS) 2025 Robot Demonstration.**
Invited to present a humanoid robot dance demonstration at GDPS, showcasing the stability and robustness of reinforcement learning policies in complex unstructured environments.
- 2026 **Open-Source Contributor, InstinctLab and GenesisLab GitHub Communities.**

Honors & Awards

- 2022 **Second Prize, RoboCup China Open – FIRA Small-Size Soccer Robot Competition, Zhengzhou University.**
 Second Prize, RoboCup China Open – 2D Soccer Simulation League, Zhengzhou University.